

Amplituhedra and Origami

Pavel Galashin (Cornell)

Combinatorics at the Confluence

July 20, 2026

[arXiv:2410.09574v2](#), [arXiv:2606.03439](#)

≡ Kawasaki's theorem

🌐 6 languages ▾

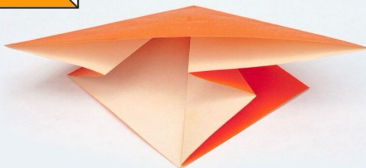
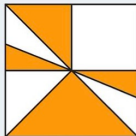
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Kawasaki's theorem or **Kawasaki–Justin theorem** is a [theorem](#) in the [mathematics of paper folding](#) that describes the [crease patterns](#) with a single [vertex](#) that may be folded to form a flat figure. It states that the pattern is flat-foldable if and only if alternately adding and subtracting the angles of consecutive folds around the vertex gives an [alternating sum](#) of zero. Crease patterns with more than one vertex do not obey such a simple criterion, and are [NP-hard](#) to fold.



In this example, the alternating sum of angles (clockwise from the bottom) is $90^\circ - 45^\circ + 22.5^\circ - 22.5^\circ + 45^\circ - 90^\circ + 22.5^\circ - 22.5^\circ = 0^\circ$. Since it adds to zero, the crease pattern may be flat-folded. 

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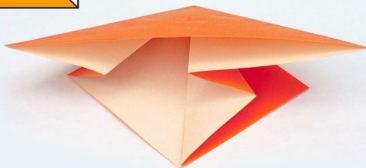
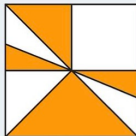
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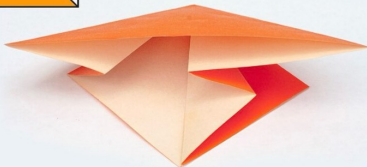
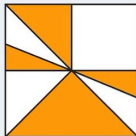
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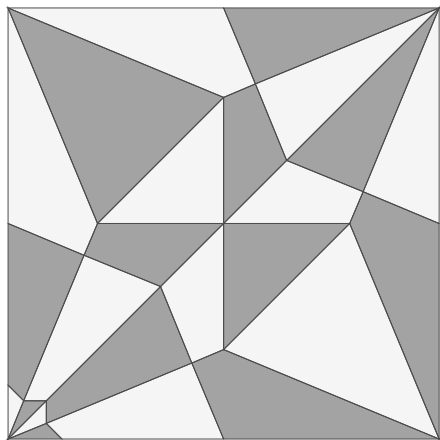


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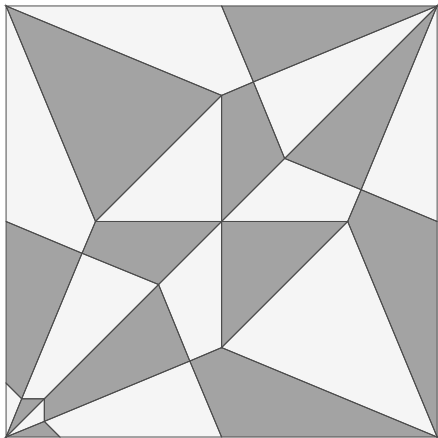


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Origami crease patterns (OCP)

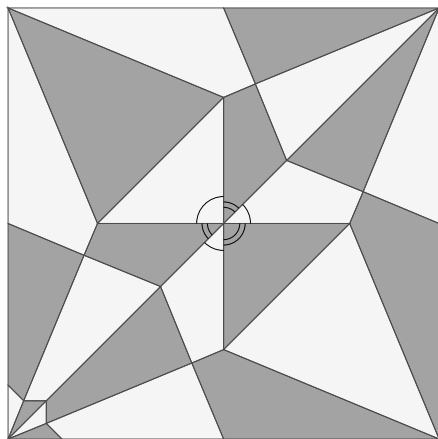


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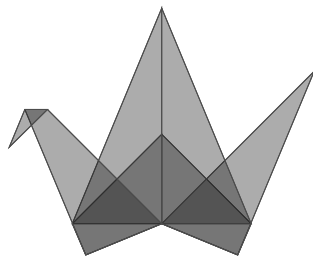
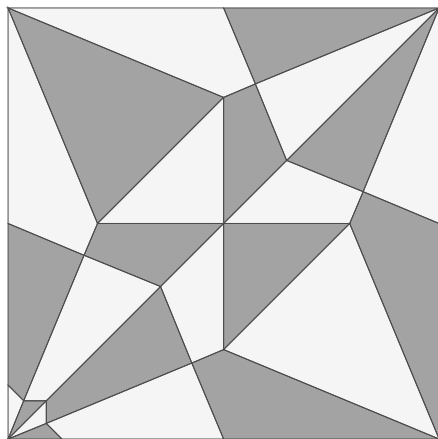
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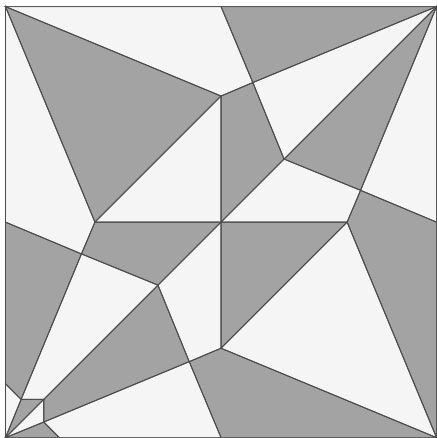


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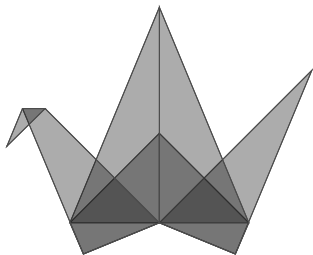
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- Faces: convex polygons colored black and white;
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- **Origami map**: isometry on each face preserving/reversing the orientations of white/black faces.

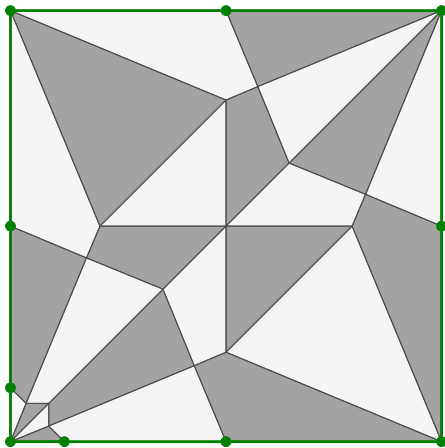


“Space” ($\mathbb{R}^2$)

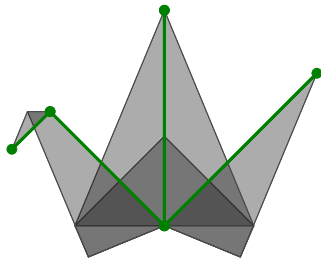


“Time” ($\mathbb{R}^2$)

- Denote boundary vertices by $\hat{x}_1, \dots, \hat{x}_n \in \mathbb{R}^2$ and $\check{x}_1, \dots, \check{x}_n \in \mathbb{R}^2$.

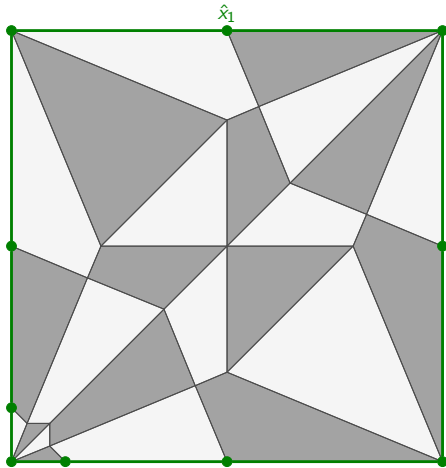


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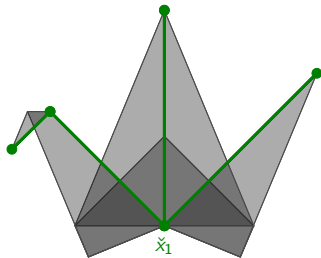


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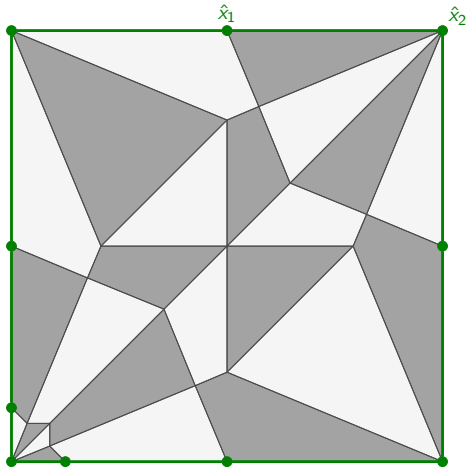


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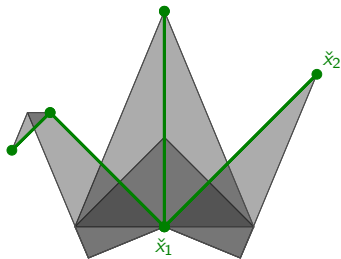


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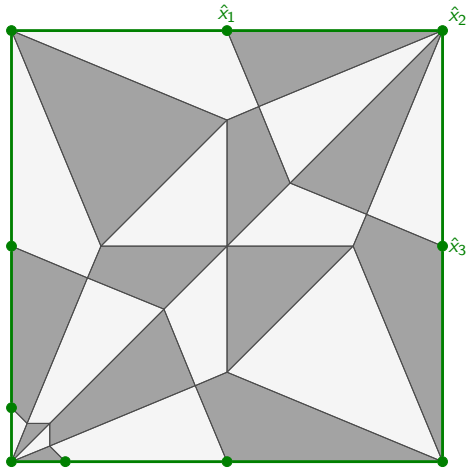


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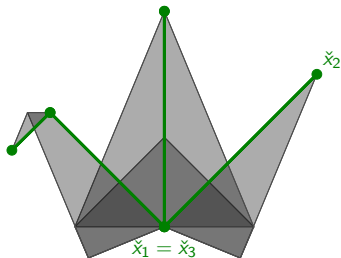


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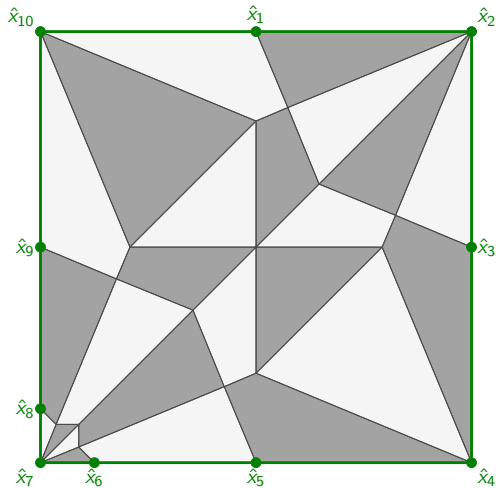


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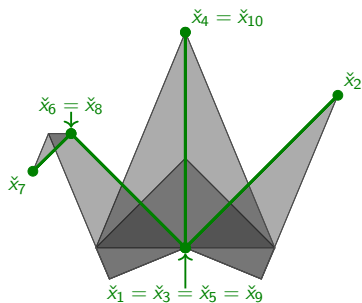


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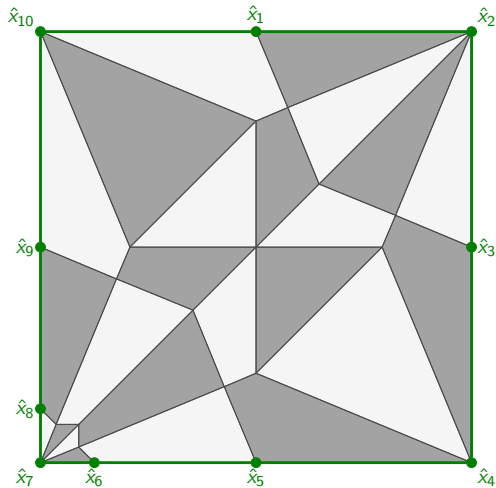


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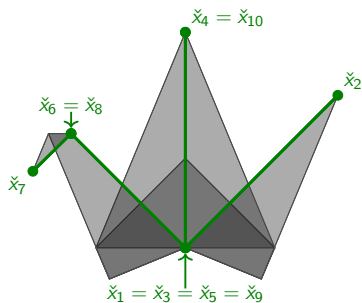


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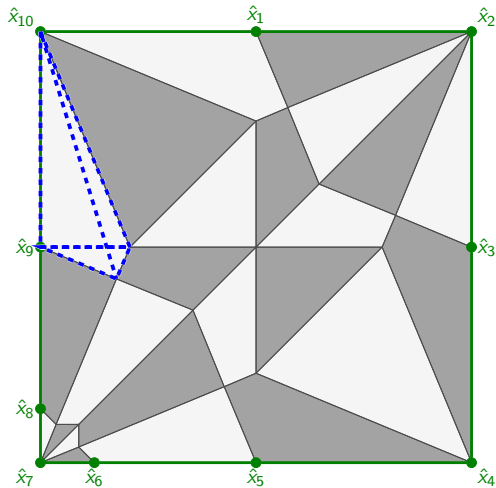


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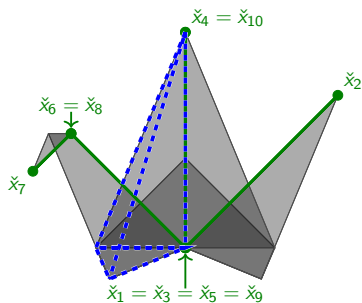


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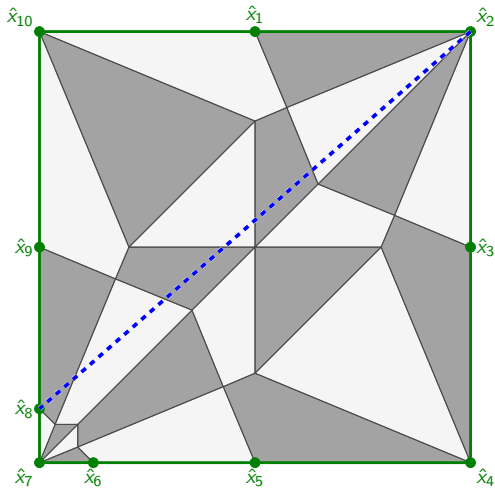


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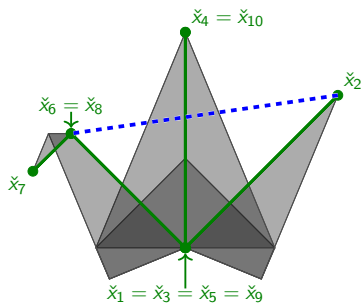


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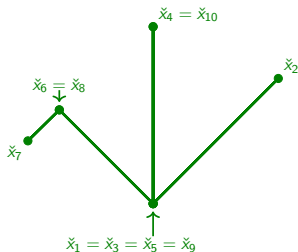
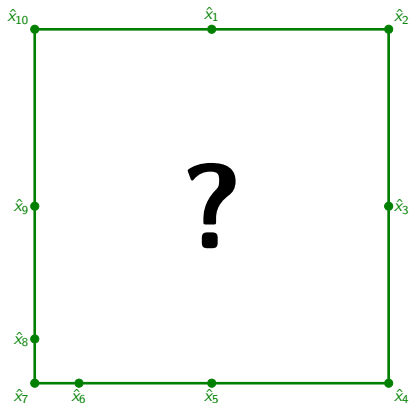


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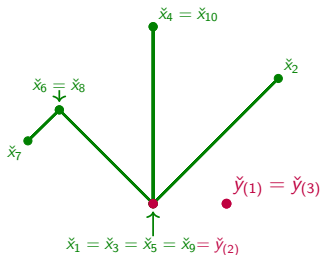
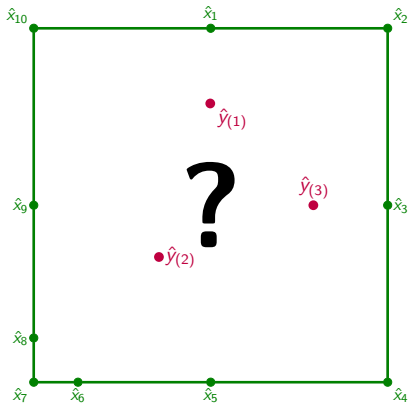
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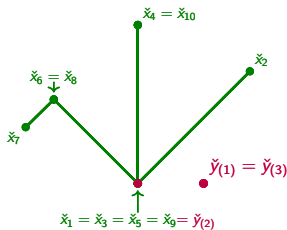
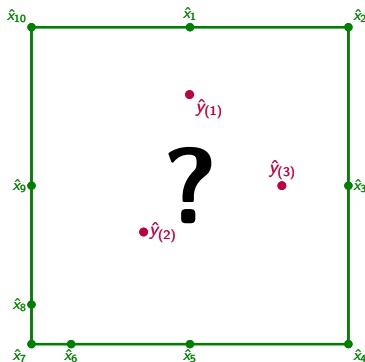
- M-positivity: origami map does not increase distances: $|\hat{x}_i - \hat{x}_j| \geq |\check{x}_i - \check{x}_j|$.
- **Origami reconstruction problem:** given points $(\hat{x}_1, \dots, \hat{x}_n)$ and $(\check{x}_1, \dots, \check{x}_n)$ satisfying $|\hat{x}_i - \hat{x}_j| \geq |\check{x}_i - \check{x}_j|$, find an OCP with boundary polygon $(\hat{x}_1, \dots, \hat{x}_n)$ whose origami map sends $\hat{x}_i \mapsto \check{x}_i$ for all i .



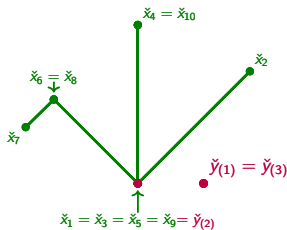
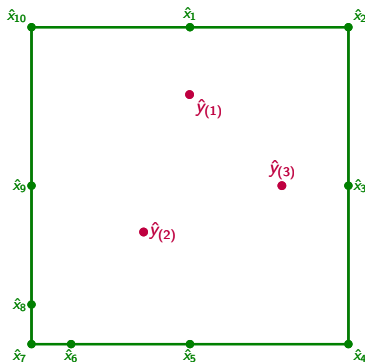
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- **L-punctured origami reconstruction problem:** same as above but we additionally require the origami map to send $(\hat{y}_{(1)}, \dots, \hat{y}_{(L)}) \mapsto (\check{y}_{(1)}, \dots, \check{y}_{(L)})$.



L-punctured origami reconstruction problem: Find an OCP with boundary $(\hat{x}_1, \dots, \hat{x}_n)$ such that the origami map sends $\hat{x}_i \mapsto \check{x}_i$ and $\hat{y}_{(j)} \mapsto \check{y}_{(j)}$ for all i, ρ .

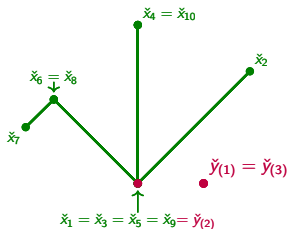
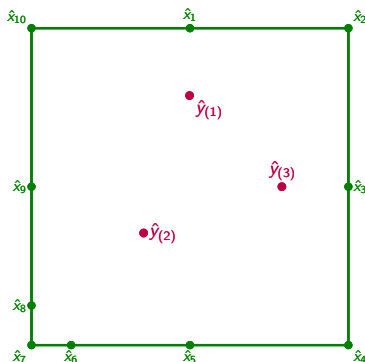


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Origami Reconstruction Algorithm

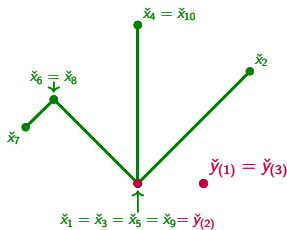
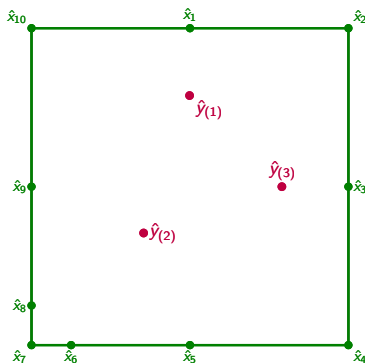
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Origami Reconstruction Algorithm

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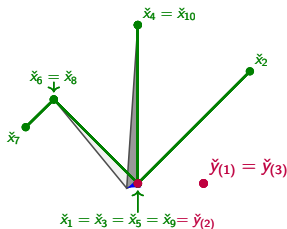
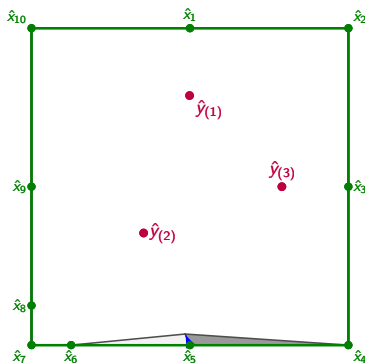
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Origami Reconstruction Algorithm

- 1 Choose $1 \leq i \leq n$ and a convention ($\circ \bullet$ vs $\bullet \circ$).
- 2 Find the unique point $\hat{x}_i(t)$ that forms white/black triangles with $\hat{x}_{i-1}, \hat{x}_i, \hat{x}_{i+1}$ and satisfies $|\hat{x}_i(t) - \hat{u}| = |\check{x}_i(t) - \check{u}|$ for exactly one other vertex $\hat{u}$.
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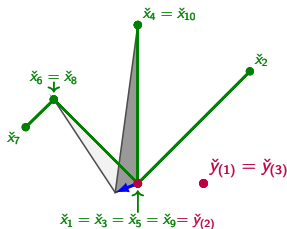
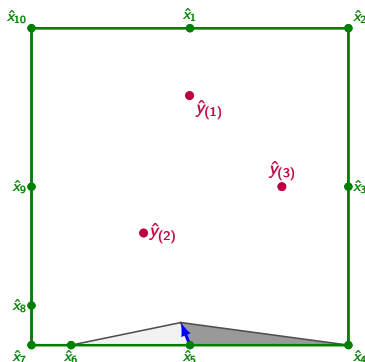
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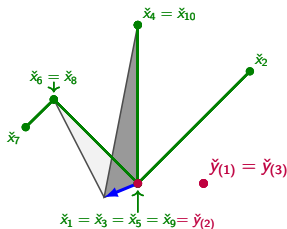
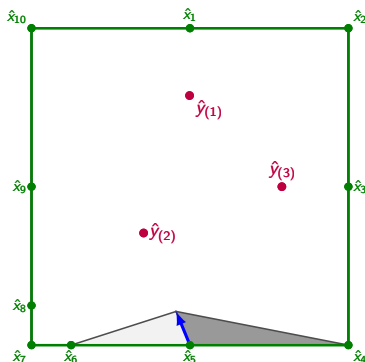
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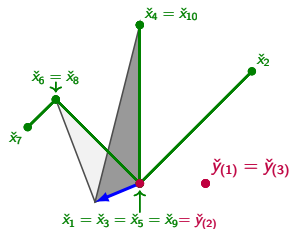
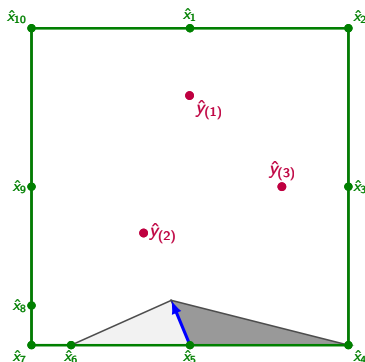
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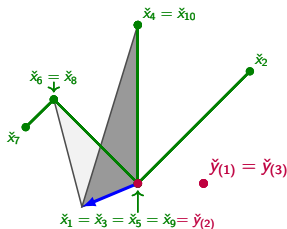
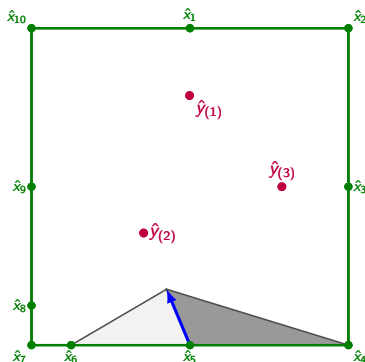
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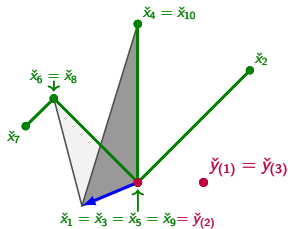
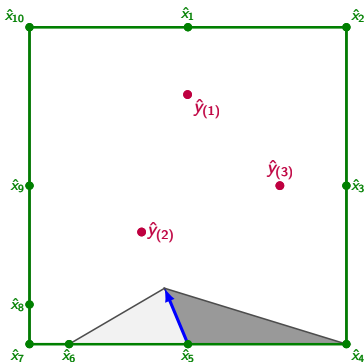
L-punctured origami reconstruction problem: Find an OCP with boundary $(\hat{x}_1, \dots, \hat{x}_n)$ such that the origami map sends $\hat{x}_i \mapsto \check{x}_i$ and $\hat{y}_{(j)} \mapsto \check{y}_{(j)}$ for all i, ρ .



Origami Reconstruction Algorithm

- 1 Choose $1 \leq i \leq n$ and a convention ($\circ\bullet$ vs $\bullet\circ$).
- 2 Find the unique point $\hat{x}_i(t)$ that forms white/black triangles with $\hat{x}_{i-1}$, $\hat{x}_i$, $\hat{x}_{i+1}$ and satisfies $|\hat{x}_i(t) - \hat{u}| = |\check{x}_i(t) - \check{u}|$ for exactly one other vertex $\hat{u}$.
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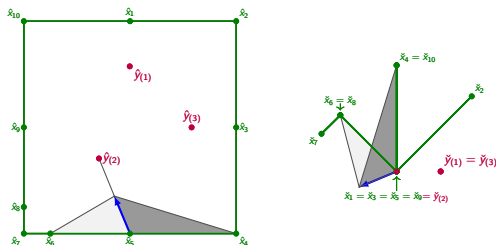
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- 3 Draw an edge from $x_i(t)$ to u and repeat.

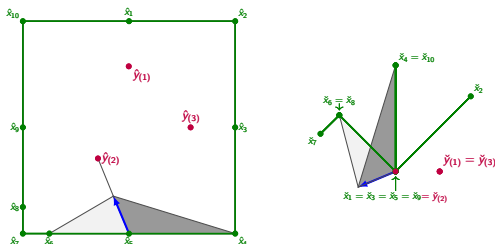
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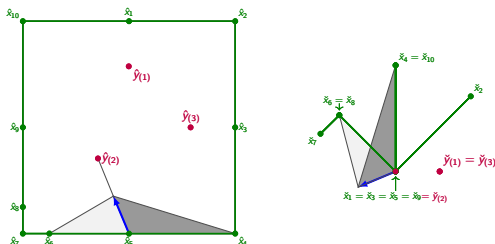
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Theorem (G. '26)

The Origami Reconstruction Algorithm *works*.

L -punctured origami reconstruction problem: Find an OCP with boundary $(\hat{x}_1, \dots, \hat{x}_n)$ such that the origami map sends $\hat{x}_i \mapsto \check{x}_i$ and $\hat{y}_{(j)} \mapsto \check{y}_{(j)}$ for all i, ρ .



Origami Reconstruction Algorithm

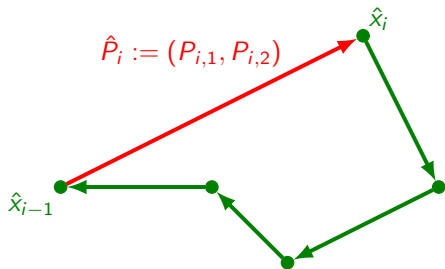
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Theorem (G. '26)

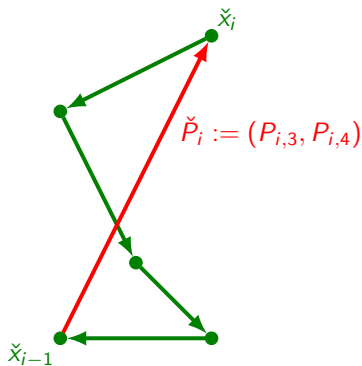
The Origami Reconstruction Algorithm works.

For all M -positive $(x_1, \dots, x_n)$, $(y_{(1)}, \dots, y_{(L)})$, it outputs a **valid, embedded OCP**.

- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, **massless:** $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$,
momentum conservation: $P_1 + P_2 + \dots + P_n = 0$.



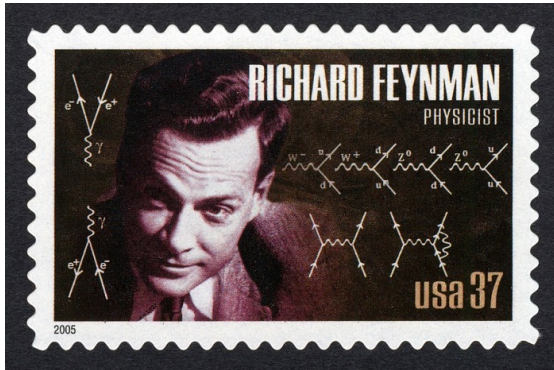
Space ($\mathbb{R}^2$)



Time ($\mathbb{R}^2$)

- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, massless: $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$,
momentum conservation: $P_1 + P_2 + \dots + P_n = 0$.
- **Output:** Scattering amplitude

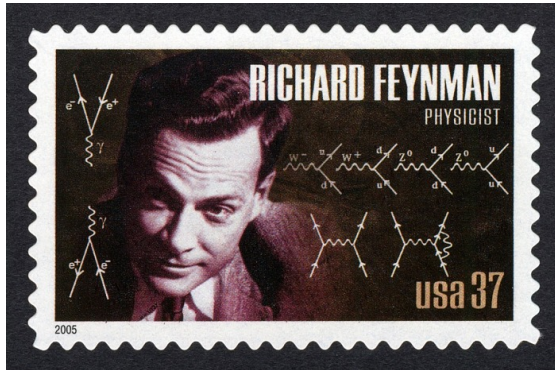
$$A(P_1, P_2, \dots, P_n) = \sum_{\text{Feynman diagram } D} A_D = A^{(L=0)} + A^{(L=1)} + A^{(L=2)} + \dots$$



- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, massless: $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$,
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$L = |E| - |V| + 1$



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- **Loop BCFW recursion:** $A^{(L)} = \sum_{\substack{\text{planar bipartite graph } \Gamma \in \Gamma_{k,n,(L)}^{\text{BCFW}} \\ \text{with } L \text{ marked faces}}} A_{\Gamma}^{(L)}.$

[Britto–Cachazo–Feng–Witten '05]

- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, massless: $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$, momentum conservation: $P_1 + P_2 + \dots + P_n = 0$.
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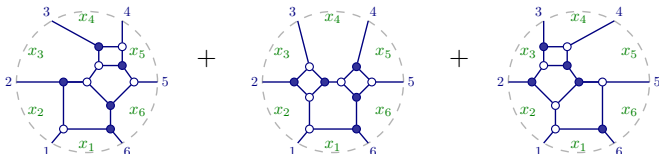
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- **Loop BCFW recursion:** $A^{(L)} = \sum \quad A_{\Gamma}^{(L)}.$

[Britto–Cachazo–Feng–Witten '05]

planar bipartite graph $\Gamma \in \Gamma_{k,n,(L)}^{\text{BCFW}}$
with L marked faces

$$\Gamma_{3,6,(0)}^{\text{BCFW}} =$$


The image shows three planar bipartite graphs, each enclosed in a dashed circle representing the boundary of the amplitude. The vertices are colored black and white in a bipartite arrangement. External legs are labeled 1 through 6. Internal edges are labeled x_1 through x_6 . The three graphs represent different ways to decompose the 3-loop amplitude into lower-loop components. The first graph has a central square face with two internal diagonals. The second graph has a central square face with one internal diagonal. The third graph has a central square face with no internal diagonals. The graphs are summed together with plus signs.

- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, massless: $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$, momentum conservation: $P_1 + P_2 + \dots + P_n = 0$.

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[Britto–Cachazo–Feng–Witten '05]

planar bipartite graph $\Gamma \in \Gamma_{k,n,(L)}^{\text{BCFW}}$
with L marked faces

$$\Gamma_{3,6,(0)}^{\text{BCFW}} =$$

+ + +

$$\Gamma_{2,4,(1)}^{\text{BCFW}} =$$

+ \dots

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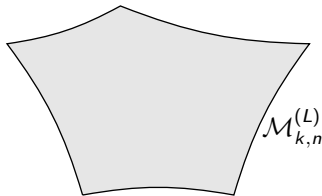
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[Britto–Cachazo–Feng–Witten '05]

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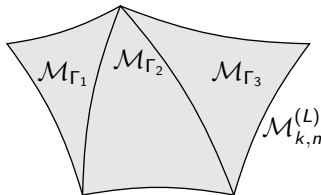
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- **Ways to run BCFW recursion** $\rightarrow$ triangulations of the amplituhedron $\mathcal{M}_{k,n}^{(L)}$.



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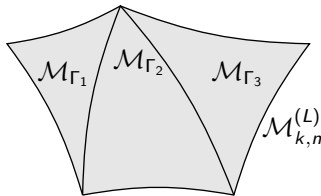
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- **Terms in BCFW recursion** $\longleftrightarrow$ “volumes” of tiles in a given triangulation.



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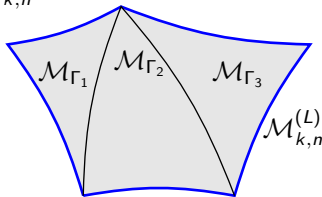
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- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, massless: $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$, momentum conservation: $P_1 + P_2 + \dots + P_n = 0$.

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[Britto–Cachazo–Feng–Witten '05]

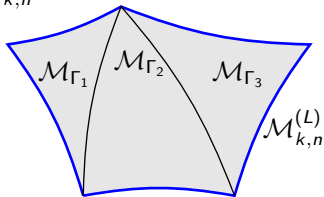
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[Britto–Cachazo–Feng–Witten '05]

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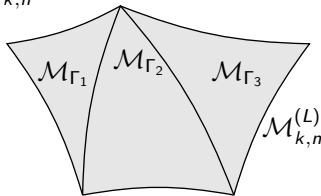
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$$(x_i - x_j)^2 = 0, (x_i - y_{(j)})^2 = 0, (y_{(j)} - y_{(k)})^2 = 0.$$

- $A^{(L)} =$ “volume” of the amplituhedron $\mathcal{M}_{k,n}^{(L)}$;



- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, massless: $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$, momentum conservation: $P_1 + P_2 + \dots + P_n = 0$.

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[Britto–Cachazo–Feng–Witten '05] planar bipartite graph $\Gamma \in \Gamma_{k,n,(L)}^{\text{BCFW}}$
with L marked faces

Amplituhedron philosophy: [Hodges '09] [Arkani-Hamed–Trnka '13]:

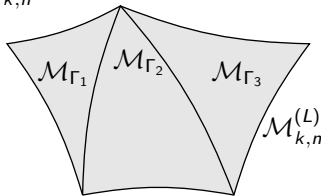
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- **Momentum amplituhedron:** $\mathcal{M}_{k,n}^{(L)}$;



- **Input:** $P_1, P_2, \dots, P_n \in \mathbb{R}^{2,2}$, massless: $P_i^2 := P_{i,1}^2 + P_{i,2}^2 - P_{i,3}^2 - P_{i,4}^2 = 0$, momentum conservation: $P_1 + P_2 + \dots + P_n = 0$.

- **Output:** Scattering amplitude

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[Britto–Cachazo–Feng–Witten '05]

planar bipartite graph $\Gamma \in \Gamma_{k,n}^{\text{BCFW}}(L)$
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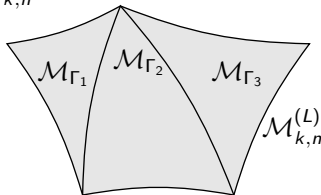
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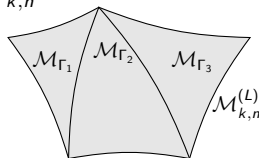
- **Momentum amplituhedron:** $\mathcal{M}_{k,n}^{(L)}$;

- **T-dual amplituhedron:** $\mathcal{A}_{k-2,n}^{(L)}$.



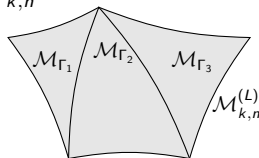
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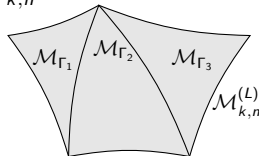


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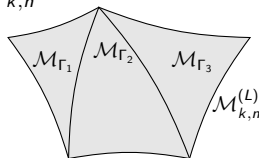
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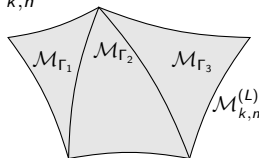
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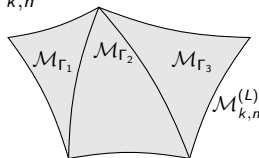
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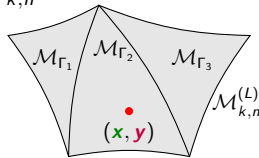
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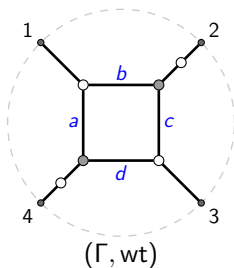
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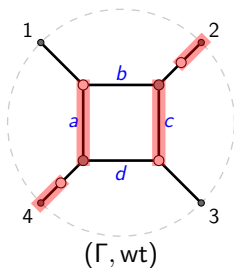


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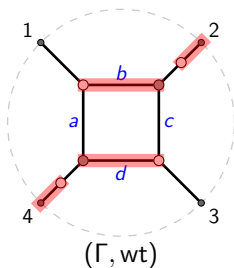


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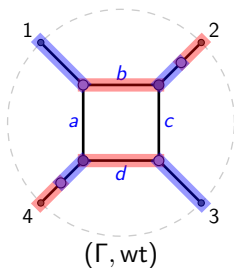


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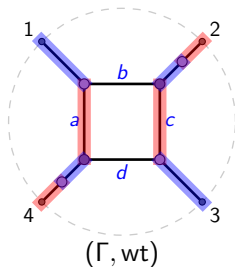
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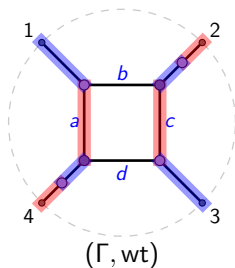
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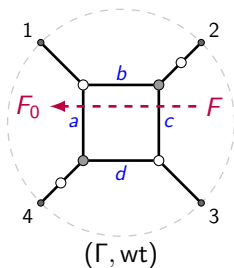
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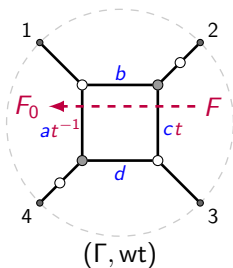
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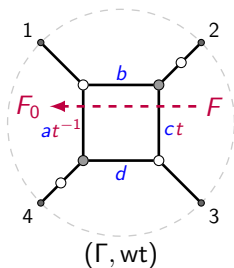
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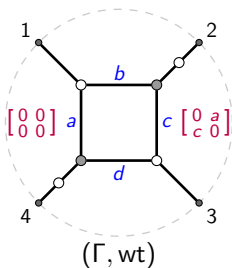
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$$C = \begin{pmatrix} 1 & b & 0 & -a \\ 0 & c & 1 & d \end{pmatrix} \in \text{Gr}_{\geq 0}(2, 4)$$

$$\Delta_{12}(C) = c \quad \Delta_{23}(C) = b \quad \Delta_{34}(C) = a \quad \Delta_{14}(C) = d$$

$$\Delta_{13}(C) = 1$$

$$\Delta_{24}(C) = ac + bd$$

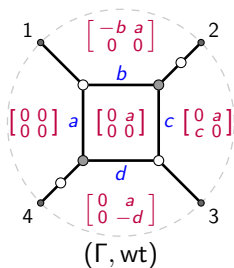
$$\text{Plücker relation: } \Delta_{13}\Delta_{24} = \Delta_{12}\Delta_{34} + \Delta_{14}\Delta_{23}$$

- Given a face F of Γ , choose a cut from F to reference face F_0 .
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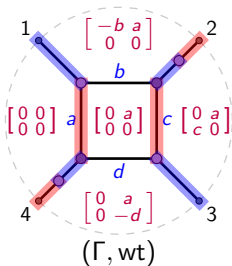
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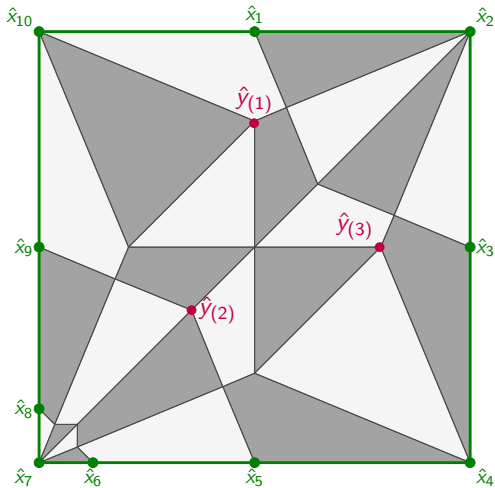
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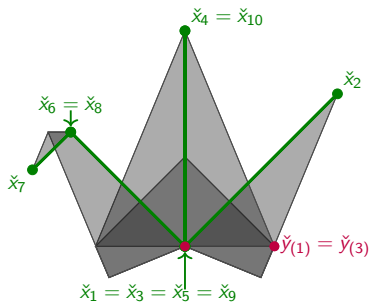
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- **M-positivity:** $\text{rank}(C'_{F_i}(1) - C'_{F_j}(1)) \geq 2 \iff F_i$ and F_j are **2-separated**
 (separated by at least 2 double-dimer paths).

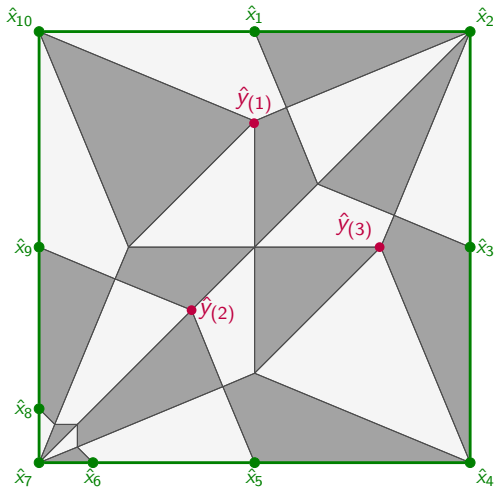


“Space” ($\mathbb{R}^2$)

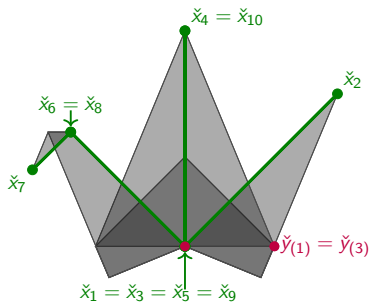


“Time” ($\mathbb{R}^2$)

- OCP + folding: embed $\Gamma^* \rightarrow \mathbb{R}^{2,2}$ with Minkowski norm $|x|^2 = |\hat{x}|^2 - |\check{x}|^2$.

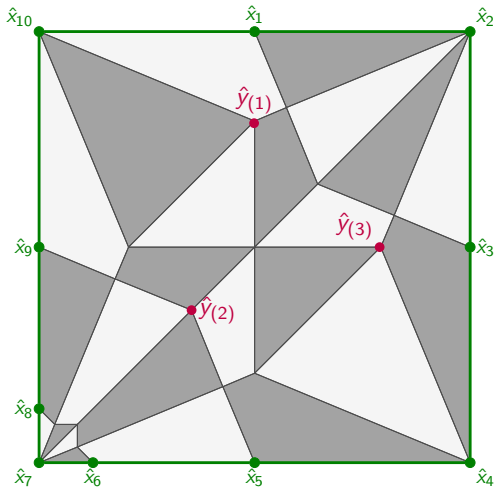


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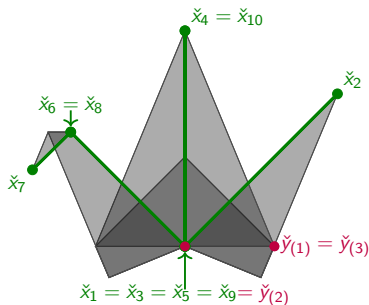


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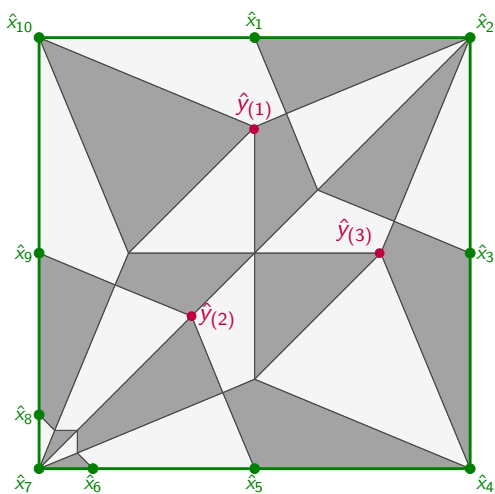


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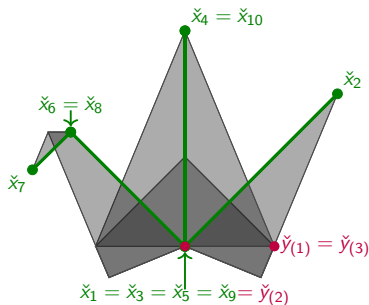


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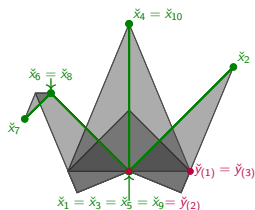
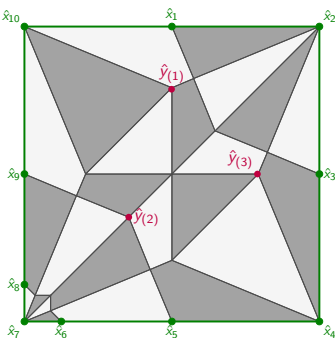


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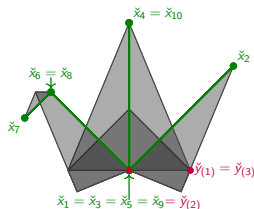
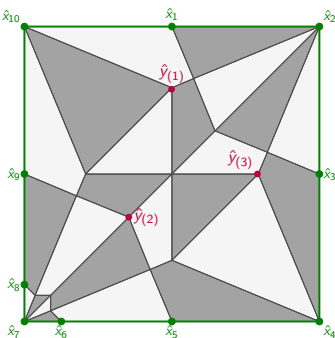


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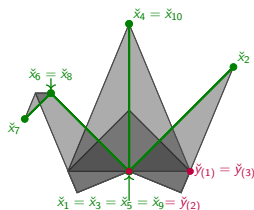
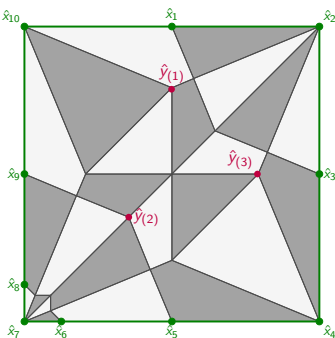
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 - Every OCP is a linear projection of a Kenyon–Smirnov primitive.
- [Kenyon–Lam–Ramassamy–Russkikh '18], [Chelkak–Laslier–Russkikh '21]



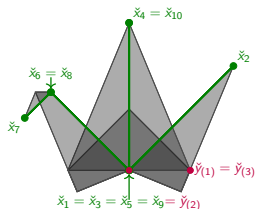
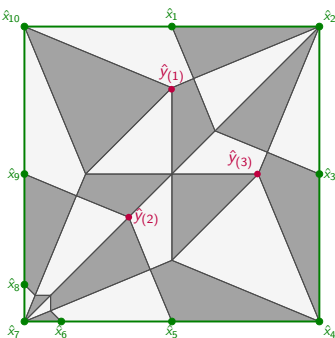
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- OCPs: **embedded** in the plane, satisfy **angle condition**: $\hat{\alpha}^\bullet = \hat{\alpha}^\circ = \pi$



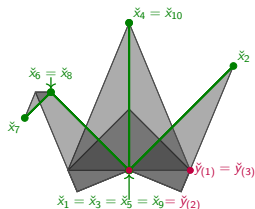
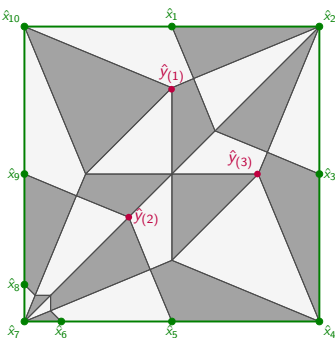
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Theorem (Origami–Amplituhedron Correspondence [G.'24])

A Kenyon–Smirnov primitive gives rise to an *embedded OCP* $\iff$

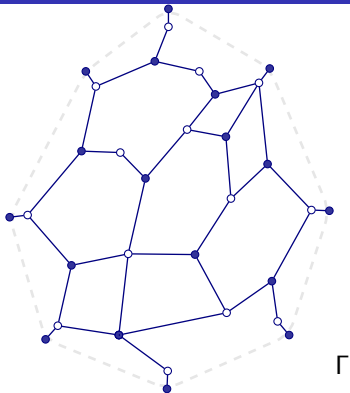


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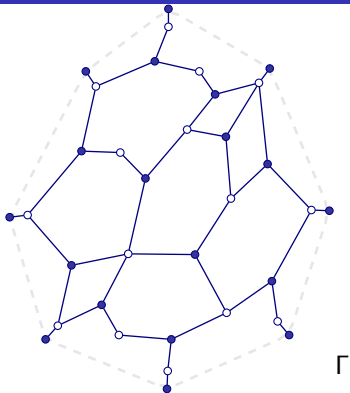
A Kenyon–Smirnov primitive gives rise to an embedded OCP $\iff$ its $\mathbb{R}^{2,2}$ -boundary polygon belongs to **momentum amplituhedron** $\mathcal{M}_{k,n}^{(L=0)}$.

T-duality for planar bipartite graphs



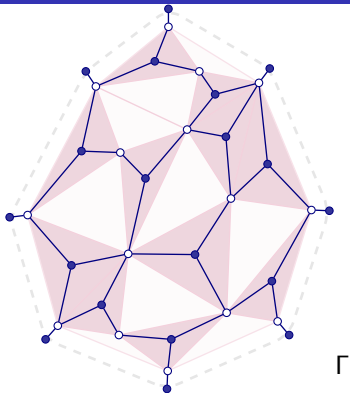
- [G. '16], [G.–Postnikov–Williams '19], [Balitskiy–Wellman '19], [Lukowski–Parisi–Williams '20], [Parisi–Sherman–Bennett–Williams '21].

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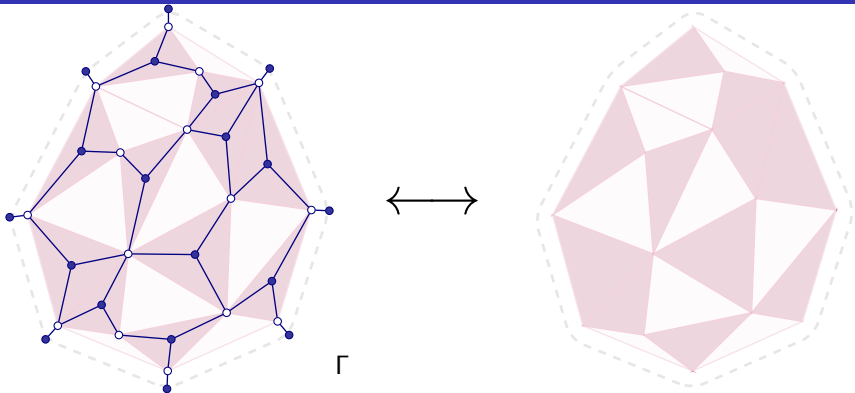
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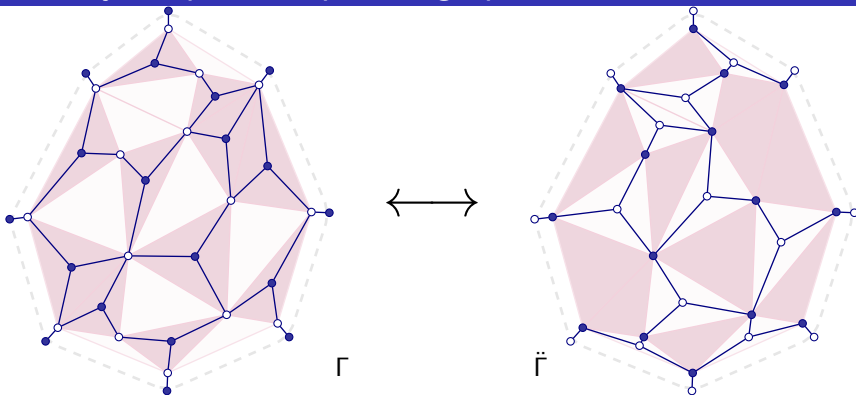
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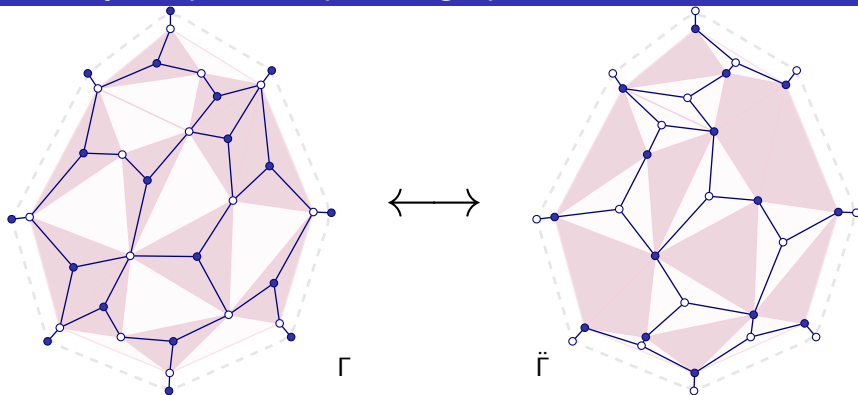
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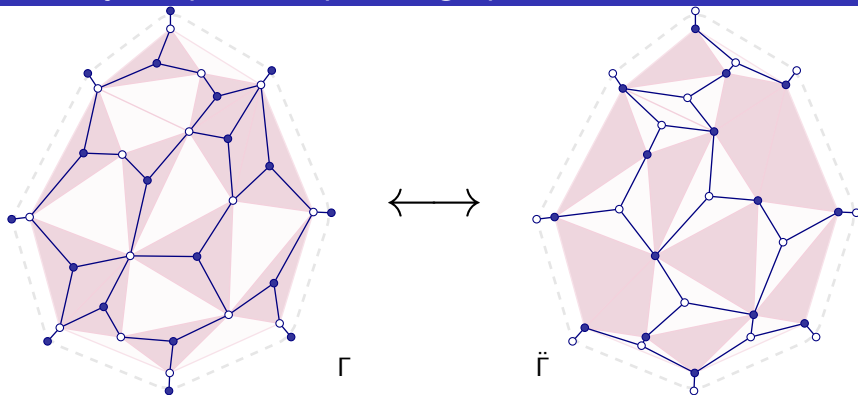
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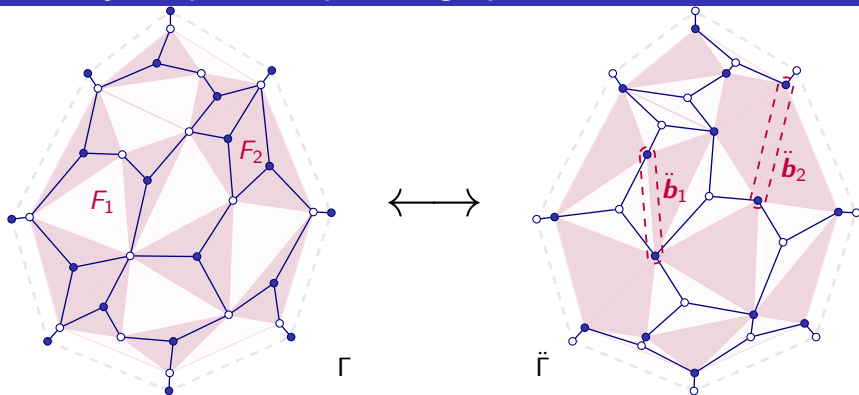
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T-duality for planar bipartite graphs

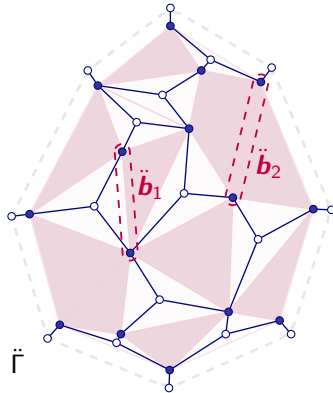
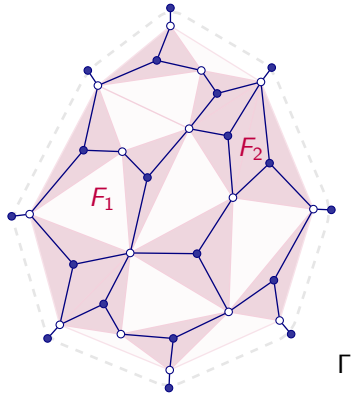


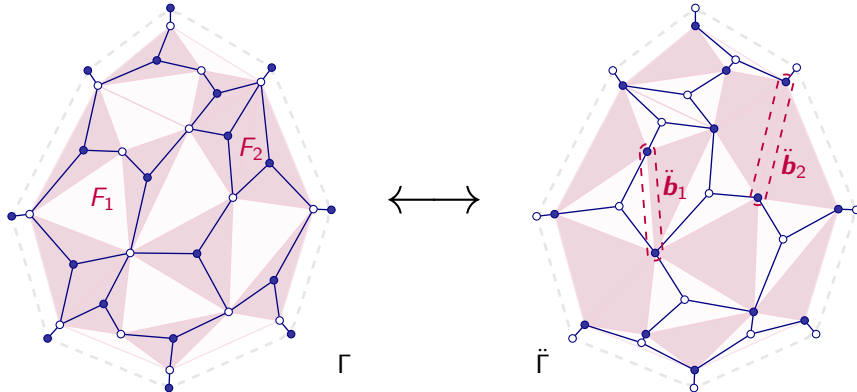
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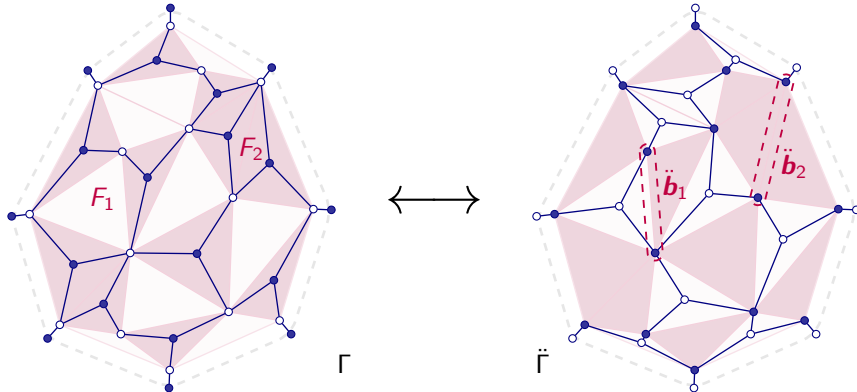
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- **Faces** of Γ correspond to **pairs of black vertices** in $\tilde{\Gamma}$.





Theorem (T-duality [G.'26])

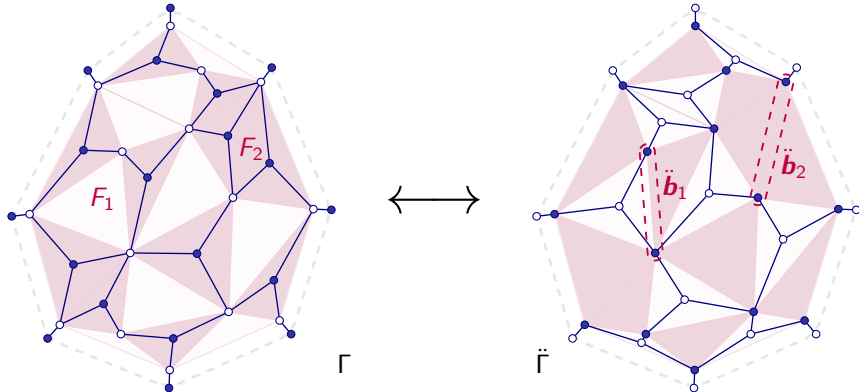
For each face F_i of Γ , let $\ddot{b}_i$ consist of any two black vertices incident to F_i .



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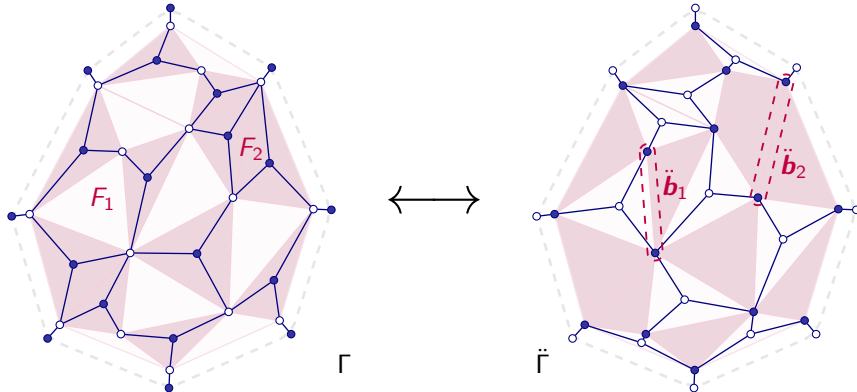
- Boundary measurements of (Γ, wt) and $(\tilde{\Gamma}, \tilde{\text{wt}})$ are related by the **magic projector** $\text{Gr}_{\geq 0}(k, n) \rightarrow \text{Gr}_{\geq 0}(k - 2, n)$ of [Arkani-Hamed–Cachazo–Cheung '10].



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- The **tangent point** $C_{F_i}^l(1)$ is related to $(\check{\Gamma} \setminus \check{b}_i, \check{\text{wt}})$ by an explicit formula.



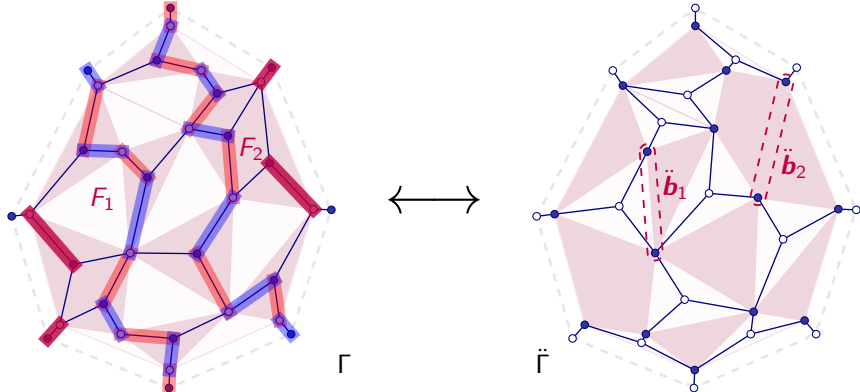
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- F_i and F_j are 2-separated $\iff \tilde{\Gamma} \setminus (\tilde{\mathbf{b}}_i \cup \tilde{\mathbf{b}}_j)$ admits a perfect matching

$\Updownarrow$
M-positivity for $\mathcal{M}_{k,n}^{(L)}$

$\Updownarrow$
T-dual M-positivity for $\mathcal{A}_{k-2,n}^{(L)}$



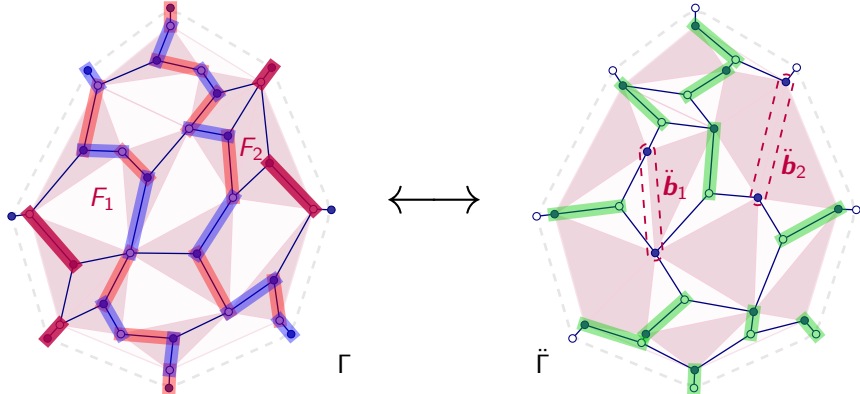
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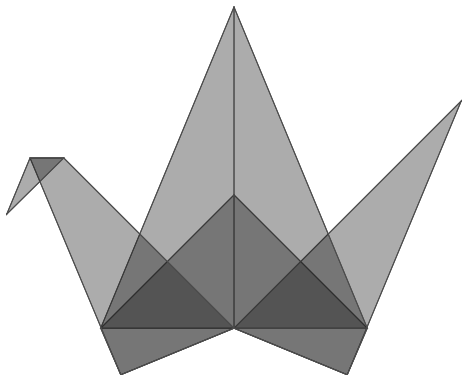
Theorem (T-duality [G.'26])

For each face F_i of Γ , let $\tilde{\mathbf{b}}_i$ consist of any two black vertices incident to F_i .

- Boundary measurements of (Γ, wt) and $(\tilde{\Gamma}, \tilde{\text{wt}})$ are related by the magic projector $\text{Gr}_{\geq 0}(k, n) \rightarrow \text{Gr}_{\geq 0}(k - 2, n)$ of [Arkani-Hamed–Cachazo–Cheung '10].
- The tangent point $C_{F_i}^L(1)$ is related to $(\tilde{\Gamma} \setminus \tilde{\mathbf{b}}_i, \tilde{\text{wt}})$ by an explicit formula.
- F_i and F_j are 2-separated $\iff \tilde{\Gamma} \setminus (\tilde{\mathbf{b}}_i \cup \tilde{\mathbf{b}}_j)$ admits a perfect matching

$\Updownarrow$
M-positivity for $\mathcal{M}_{k,n}^{(L)}$

$\Updownarrow$
T-dual M-positivity for $\mathcal{A}_{k-2,n}^{(L)}$



Thanks!